Superconducting films

X21 (2021) — English translation

In thin films $\mathrm{In-O}$ at low temperatures (of the order of 1 K), a superconductor-insulator phase transition controlled by an external magnetic field is observed (the transition occurs when the external magnetic field B changes).

Part A. Critical parameters (6.8 points)

The table "Experimental data" presents measurements of the dependences $R(T)$, where R is the resistance of the sample, T is the temperature, at different external magnetic fields B. Resistance measurement error $\varepsilon(R) = 0.5\%$. The graph (see Fig.) shows approximate types of dependencies $R(T)$ for some values B indicated in the table. Let us assume that the dependence of resistance $R(T, B)$ has the following form:

$$R(T,B) = R_C - \gamma T + A\left(B - B_C\right)T^{\alpha},$$

where A, R_C, B_C — , . on $R(T)$ (then $\dfrac{\partial R}{\partial T} = 0$), then , $(T_\mathrm{n}, R_\mathrm{n})$. : h : , for h x y constant and vice versa, for example:

$$f(x,y) = 3x^2 + 2xy, \quad \frac{\partial}{\partial x} f(x,y) = 6x + 2y, \quad \frac{\partial}{\partial y} f(x,y) = 2x$$

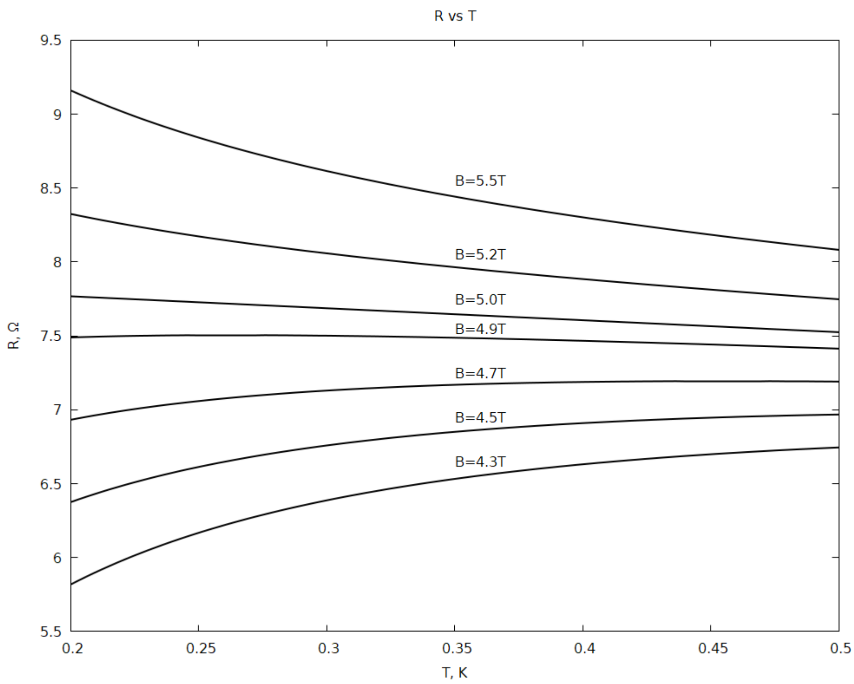

Figure 1: Figure 1.

A1	Derive the analytical relationship $R_\mathrm{p}\ (T_\mathrm{p})$.	**1.00**

A2	Define B_C.	**0.40**

A3	Define R_C in two ways.	**2.60**

A4	Define the parameters γ and α.	**2.80**

Part B. Probability of describing an experiment (3.2 points)

Estimating the error of a physical quantity X allows one to calculate the probability with which its value lies in the interval $[a, b]$. To assess the probability p that the result of the experiment is described by a mathematical expression, the criterion χ^2 (chi-square) is used, calculated according to the following formula:

$$\chi^2 = \sum_{i=1}^{N} \left(\frac{R - R}{\Delta R} \right)^2$$

For a number of points $N = 150$, a correspondence table p and χ^2:
(Probability table)

B1	Count χ^2 (definition in note) of $R(T)$ dependencies for each B.	**2.20**

B2	Using the table from the note, estimate the probability according to which the formula $f = R(T, B)$ describes the dependence $R(T, B)$ for any (specify which) 150 points.	**1.00**

Source: `https://pho.rs/p/102`